Chapter 8, Lecture 6

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Large Sample Theory

We have seen that quite often, but not always, the MLE is consistent and asymptotically normal $N\left(\theta, 1/\{nI(\theta)\}\right)$ for large values of the sample size n. Here the *Fisher information* $I(\theta)$ is given by

$$\mathbb{E}_\theta \left(\frac{\partial}{\partial \theta} \log f(X|\theta) \right)^2 .$$

A useful formula is

$$\mathbb{E}_\theta \left(\frac{\partial}{\partial \theta} \log f(X|\theta) \right)^2 = -\mathbb{E}_\theta \left(\frac{\partial^2}{\partial \theta^2} \log f(X|\theta) \right),$$

especially since the term on the right is often easier to compute.

We will discuss the regular case where we have n iid observations from a "nice" pdf $f(x|\theta_0)$. Nice means here that $\theta \to f(x|\theta)$ is smooth enough and the domain of $f(x|\theta)$ does not depend on θ. 1. Note that by the LLN, we have

$$\frac{1}{n}\ell(\theta) = \frac{1}{n}\sum_{i=1}^n \log f(X_i|\theta) \xrightarrow{P} \mathbb{E}_{\theta_0} \log f(X|\theta)]$$

The true value θ_0 is a stationary point of the RHS under suitable regularity as

$$\begin{aligned}
\frac{\partial}{\partial \theta}\mathbb{E}_{\theta_0} \log f(X|\theta) &= \int \frac{\dot{f}(x|\theta)}{f(x|\theta)} f(x|\theta_0)\,\mathrm{d}x \\
&= \int \dot{f}(x|\theta_0)\,\mathrm{d}x \qquad \text{for } \theta = \theta_0 \\
&= \frac{\partial}{\partial \theta} 1 \\
&= 0
\end{aligned}$$

Although it is plausible that the maximizer $\widehat{\theta}$ of $\ell(\theta)$ is close to that of the maximizer θ_0 of its expected value $\mathbb{E}_{\theta_0} \log f(X|\theta)]$ in view of the convergence

$$\frac{1}{n}\ell(\theta) \xrightarrow{P} \mathbb{E}_{\theta_0} \log f(X|\theta)],$$

this in itself does not guarantee that $\widehat{\theta}$ is consistent, as the convergence holds for each θ, as $n \to \infty$, and not necessary uniform in θ. Remember that $\widehat{\theta}$ is random.

2. We denote by

$$s(x|\theta) = \frac{\partial}{\partial\theta} \log f(x|\theta)$$

the score function. Typically, it has mean zero,

$$\mathbb{E}_\theta[s(X|\theta)] = \int \frac{\dot{f}(x|\theta)}{f(x|\theta)} f(x|\theta)\,\mathrm{d}x = \int \dot{f}(x|\theta)\,dx = \frac{\partial}{\partial\theta}\int f(x|\theta)\,dx = 0$$

and we denote its variance by

$$I(\theta) = \mathrm{Var}(s(X|\theta)) = \mathbb{E}[s^2(X|\theta)].$$

Note that

$$\begin{aligned} 0 &= \frac{\partial}{\partial\theta}\mathbb{E}[s(X|\theta)] = \frac{\partial}{\partial\theta}\int s(x|\theta) f(x|\theta)\mathrm{d}x \\ &= \int \dot{s}(x|\theta) f(x|\theta)\mathrm{d}x + \int s(x|\theta)\dot{f}(x|\theta)\mathrm{d}x \\ &= \int \left(\frac{d^2}{d\theta^2}\log f(x|\theta)\right) f(x|\theta)\mathrm{d}x + \int s^2(x|\theta) f(x|\theta)\mathrm{d}x \end{aligned}$$

which implies the formula

$$\begin{aligned} I(\theta) &= \mathbb{E}_\theta \left(\frac{\partial}{\partial\theta}\log f(X|\theta)\right)^2 \\ &= -\mathbb{E}_\theta \left(\frac{\partial^2}{\partial\theta^2}\log f(X|\theta)\right). \end{aligned}$$

3. Provided $\widehat{\theta} \xrightarrow{P} \theta_0$, we can use the Taylor expansion

$$0 \;=\; \ell'(\widehat{\theta}) = \ell'(\theta_0) + (\widehat{\theta} - \theta_0)\ell''(\theta_0) + \cdots$$

By the LLN, we have

$$\frac{1}{n}\ell''(\theta_0) \xrightarrow{P} -I(\theta_0)$$

and

$$\frac{1}{n}\ell'(\theta_0) \approx N(0, I(\theta_0)/n) \qquad \text{in distribution, by the CLT}$$

Combining this, gives

$$\widehat{\theta} - \theta_0 \approx -\frac{\ell'(\theta_0)/n}{\ell''(\theta_0)/n} \approx N(0, 1/(nI(\theta_0)) \qquad \text{in distribution}$$

and the conclusion

$$\widehat{\theta} \approx N(\theta_0, 1/(nI(\theta_0))$$

follows.

Notice that

$$\ell'(\theta) = s(\mathbf{X}|\theta) = \sum_{i=1}^{n} s(X_i|\theta)$$

and that

$$I_n(\theta) := \mathrm{Var}(s(\mathbf{X}|\theta)) = \sum_{i=1}^{n} \mathrm{Var}(s(X_i|\theta)) = nI(\theta).$$

Let's give some examples and a counter-example.

Exponential(λ)

Recall

$$f(x|\lambda) = \lambda \exp(-\lambda x),\ x \geq 0$$

We will compute the score function $s(x|\theta)$, the Fisher information $I(\theta)$ in two different ways, and compare the asymptotic variance $1/(nI(\theta))$ with the variance we had derived before. Verify that

$$\begin{aligned} s(x|\lambda) &= \frac{\partial}{\partial \lambda} \log f(x|\lambda) \\ &= \frac{\partial}{\partial \lambda} (\log \lambda - \lambda x) \\ &= \frac{1}{\lambda} - x. \end{aligned}$$

We see that the score function has mean zero,

$$\mathbb{E}[s(X|\lambda)] = \mathbb{E}\left[\frac{1}{\lambda} - X\right] = 0$$

The Fisher information equals on the one hand

$$I(\lambda) = \mathbb{E}\left[\left(\frac{1}{\lambda} - X\right)^2\right] = \mathrm{Var}(X) = \frac{1}{\lambda^2}$$

and on the other hand,

$$I(\lambda) = -\mathbb{E}\left[\frac{\partial}{\partial \lambda} s(X|\lambda)\right] = -\mathbb{E}\left[-\frac{1}{\lambda^2}\right] = \frac{1}{\lambda^2}$$

The formula

$$\frac{1}{nI(\lambda)} = \frac{\lambda^2}{n}$$

for the asymptotic variance of $\widehat{\lambda}$ agrees with the approximate variance we found using the delta-method earlier, and is close to the exact variance

$$\mathrm{Var}(\widehat{\lambda}) = \frac{n^2}{(n-1)^2} \frac{\lambda^2}{n-2}$$

Poisson(μ)

Here the score function equals

$$s(x|\mu) = \frac{\partial}{\partial \mu} \log\left(\mu^x \exp(-\mu)/x!\right) = \frac{x}{\mu} - 1$$

Again,

$$\mathbb{E}[s(X|\mu)] = \mathbb{E}\left[\frac{X}{\mu} - 1\right] = \frac{\mu}{\mu} - 1 = 0$$

and

$$I(\mu) = \mathbb{E}[s^2(X|\mu)] = \frac{1}{\mu^2}\mathbb{E}[(X-\mu)^2] = \frac{1}{\mu}$$

or using

$$I(\mu) = -\mathbb{E}\left[\frac{\partial}{\partial \mu} s(X|\mu)\right] = -\mathbb{E}\left[-\frac{X}{\mu^2}\right] = \frac{1}{\mu}$$

The MLE of μ is

$$\widehat{\mu} = \overline{X}_n$$

which has mean μ and variance μ/n. This coincides with the formula $1/(nI(\mu))$.

Hardy-Weinberg

This is a somewhat confusing case as we have one single observation (x_1, x_2, x_3) from a multinomial distribution (that, yes, is based on n independent trials). Recall that

$$\widehat{\theta} = \frac{X_2 + 2X_3}{2n}$$

and that we had computed

$$s(\mathbf{X}|\theta) = \ell'(\theta) = -\frac{2X_1 + X_2}{1-\theta} + \frac{X_2 + 2X_3}{\theta}$$

so that

$$\begin{aligned}
\mathbb{E}[s(\mathbf{X}|\theta)] &= -\frac{2\mathbb{E}[X_1]+\mathbb{E}[X_2]}{1-\theta}+\frac{\mathbb{E}[X_2]+2\mathbb{E}[X_3]}{\theta} \\
&= -\frac{2n(1-\theta)^2+2n\theta(1-\theta)}{1-\theta}+\frac{2n\theta(1-\theta)+2n\theta^2}{\theta} \\
&= -2n(1-\theta)-2n\theta+2n(1-\theta)+2n\theta \\
&= 0
\end{aligned}$$

Furthermore,

$$\frac{\partial}{\partial\theta}s(X|\theta)=\ell''(\theta)=-\frac{2X_1+X_2}{(1-\theta)^2}-\frac{X_2+2X_3}{\theta^2}$$

whence

$$\begin{aligned}
I(\theta) &= -\mathbb{E}\left[-\frac{2X_1+X_2}{(1-\theta)^2}-\frac{X_2+2X_3}{\theta^2}\right] \\
&= \frac{2n(1-\theta)^2+2n\theta(1-\theta)}{(1-\theta)^2}+\frac{2n\theta(1-\theta)+2n\theta^2}{\theta^2} \\
&= 2n\left\{1+\frac{\theta}{1-\theta}+\frac{1-\theta}{\theta}+1\right\} \\
&= 2n\left\{2+\frac{\theta}{1-\theta}+\frac{1-\theta}{\theta}\right\} \\
&= 2n\left\{2-1+\frac{1}{1-\theta}+\frac{1}{\theta}-1\right\} \\
&= \frac{2n}{\theta(1-\theta)}
\end{aligned}$$

and we see that the exact variance $\theta(1-\theta)/(2n)$ of $\widehat{\theta}$ is the same as $1/I(\theta)$ (remember, only one observation!).

NORMAL(μ, σ^2)

Note that

$$\begin{aligned}\frac{\partial}{\partial\mu} f(x|\mu,\sigma) &= \frac{\partial}{\partial\mu}\left\{-\frac{1}{2}\log(2\pi) - \log(\sigma) - \frac{1}{2\sigma^2}(x-\mu)^2\right\} \\ &= \frac{x-\mu}{\sigma^2}\end{aligned}$$

We see that indeed $\mathbb{E}[X-\mu]/\sigma^2 = 0$ and

$$I(\mu) = \sigma^{-4}\mathbb{E}[X-\mu]^2 = \sigma^{-2}$$

hence the asymptotic variance equals σ^2/n. This agrees with the exact variance of the MLE $\widehat{\mu} = \overline{X}$. Furthermore,

$$\begin{aligned}\frac{\partial}{\partial\sigma}\log f(x|\mu,\sigma) &= \frac{\partial}{\partial\mu}\left\{-\frac{1}{2}\log(2\pi) - \log(\sigma) - \frac{1}{2\sigma^2}(x-\mu)^2\right\} \\ &= -\frac{1}{\sigma} + \frac{(x-\mu)^2}{\sigma^3}\end{aligned}$$

Again, we note that

$$\mathbb{E}\left[-\frac{1}{\sigma} + \frac{(X-\mu)^2}{\sigma^3}\right] = -\frac{1}{\sigma} + \frac{\sigma^2}{\sigma^3} = 0$$

and

$$I(\sigma) = -\mathbb{E}\left[\frac{\partial}{\partial\sigma}\left\{-\frac{1}{\sigma} + \frac{(X-\mu)^2}{\sigma^3}\right\}\right] = -\frac{1}{\sigma^2} + \frac{3}{\sigma^2} = \frac{2}{\sigma^2}$$

It is not straightforward to compute variance of the MLE $\widehat{\sigma}$ of σ. If we reparametrize, and let $\theta = \sigma^2$, then we have that

$$I(\theta) = 1/(2\theta^2) = 1/(2\sigma^4)$$

and

$$\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

We leave this as an exercise. Soon (see lecture 9), we will find that $n\widehat{\theta}/\sigma^2$ has a known distribution (a so-called chi-square distribution with $n-1$ degrees of freedom) with variance $2(n-1)$. This means that

$$2(n-1) = \text{Var}(n\widehat{\theta}/\sigma^2) = \frac{n^2}{\sigma^4}\text{Var}(\widehat{\theta})$$

or

$$\text{Var}(\widehat{\theta}) = \frac{2\sigma^4}{n}\frac{n-1}{n}$$

which is almost the same as the asymptotic variance formula $1/(nI(\theta)) = 2\sigma^4/n$.

Counter Example: Uniform$(0,\theta)$

We found that the MLE $\widehat{\theta}$ equals $X_{(n)} = \max(X_1, \ldots, X_n)$. We already saw that the MSE had different behavior. Its asymptotic sampling distribution also differs dramatically. Since

$$\begin{aligned}
\mathbb{P}\{n(\theta - \widehat{\theta}) \le x\} &= 1 - \mathbb{P}\{\widehat{\theta} < \theta - x/n\} \\
&= 1 - \mathbb{P}\{X_1 < \theta - x/n, \cdots, X_n < \theta - x/n\} \\
&= 1 - \mathbb{P}\{X_1 < \theta - x/n\} \cdots \mathbb{P}\{X_n < \theta - x/n\} \\
&= 1 - \left(\frac{\theta - x/n}{\theta}\right)^n \\
&\to 1 - \exp(x/\theta),
\end{aligned}$$

we see that

$$n(\widehat{\theta} - \theta) \approx \text{Exponential}(1/\theta) \qquad \text{in distribution.}$$

The rate differs (n versus $\sqrt{n}$) as well as the distribution (Exponential versus Normal).